

SSH Authentication Log Analysis

Threat Hunting and Brute Force Detection using Splunk

Prepared By	Majed Almabadi
Date	19/06/2026
Tool	Splunk Enterprise - Local Instance
Dataset	ssh_logs.json
Source Type	JSON
Project Type	Authentication Log Analysis

Purpose

This report documents an SSH log investigation performed in Splunk. The analysis focuses on authentication outcomes, repeated failed attempts, suspicious source hosts, affected destination servers, and practical detection logic.

1. Executive Summary

This project analyzes SSH authentication events using Splunk. The objective is to identify abnormal login behavior and potential brute-force activity from JSON-based SSH logs.

Dataset	ssh_logs.json
Index	auth_logs
Main Fields	ts, uid, id.orig_h, id.resp_h, auth_success, auth_attempts, event_type
Total Events	2,400
Failed Authentication Events	1216
Successful Authentication Events	612
Top Suspicious Source IP	10.0.0.25
Main Finding	Source IP 10.0.0.25 showed repeated failed SSH authentication activity, including multiple failed authentication attempts with a maximum auth_attempts value of 8.

2. Environment and Dataset

2.1 Lab Environment

SIEM Tool	Splunk Enterprise
Deployment	Local Splunk instance
Operating System	Ubuntu 26.04 LTS
Access Method	Splunk Web Interface
Data Source	JSON-formatted Zeek-style SSH log file from 30-Days SOC Challenge Beginner - Day #18

2.2 Dataset Fields

Field	Description
ts	Timestamp of the event
uid	Unique connection/session identifier
id.orig_h	Source host IP address
id.orig_p	Source port
id.resp_h	Destination server IP address
id.resp_p	Destination port, usually 22 for SSH
proto	Transport protocol
auth_success	Authentication outcome: true, false, or null
auth_attempts	Number of authentication attempts
event_type	Human-readable event classification

3. Investigation Methodology

1. Data Ingestion	Upload the JSON SSH log file and verify events are searchable.
2. Field Discovery	Confirm field extraction for source IP, destination IP, authentication result, attempts, and event type.
3. Baseline Activity	Count event types and authentication outcomes to understand normal and abnormal patterns.
4. Suspicious Source Identification	Identify source hosts with repeated failed authentication attempts.
5. Timeline Reconstruction	Review suspicious source activity chronologically.
6. Detection Logic	Create an SPL search that can be converted into a Splunk alert.

4. SPL Investigation Queries

Query 1: Verify Data Ingestion

Objective: Confirm that the uploaded SSH log events are searchable in the selected index.

```
index=auth_logs
| stats count
```

Evidence:

The dataset was successfully ingested into the auth_logs index. Splunk returned 2400 events.

count
2400

Query 2: Event Type Breakdown

Objective: Identify the available event categories and their frequency.

```
index=auth_logs
| stats count by event_type
| sort -count
```

Evidence:

The event_type field shows the main SSH activity categories in the dataset, including successful logins, failed logins, multiple failed authentication attempts, and connections without authentication.

Query 3: Authentication Outcome Distribution

Objective: Compare successful, failed, and unauthenticated connection events.

```
index=auth_logs
| stats count by auth_success
```

Evidence:

The auth_success field was used to separate successful authentication events, failed authentication events, and connections where authentication was not completed.

auth_success	count
false	1216
null	572
true	612

Query 4: Top Source IP Addresses

Objective: Find the most active source hosts generating SSH activity.

```
index=auth_logs
| stats count by id.orig_h
| sort -count
```

Evidence:
This query identifies the most active source IP addresses generating SSH-related activity. The top source IPs were reviewed to determine whether any host showed abnormal authentication behavior.

id.orig_h ↕	count ↕
10.0.0.25	78
10.0.0.18	64
10.0.0.21	64
10.0.0.48	64
10.0.0.14	62
10.0.0.44	62
10.0.0.57	62
10.0.0.16	60

The Most active IP is 10.0.0.25 with 78

Query 5: Top Destination Servers

Objective: Identify which SSH servers received the most authentication activity.

```
index=auth_logs
| stats count by id.resp_h
| sort -count
```

Evidence:
This query identifies the destination SSH servers that received the highest number of events. These hosts were reviewed as potentially targeted systems.

id.resp_h ↕	count ↕
10.0.1.2	230
10.0.1.6	230
10.0.1.9	226
10.0.1.4	218
10.0.1.10	208
10.0.1.1	198
10.0.1.12	198
10.0.1.7	186

Query 6: Brute Force Candidates

Objective: Isolate failed authentication events with repeated attempts.

```
index=auth_logs auth_success=false
| where auth_attempts >= 5
| table _time uid id.orig_h id.resp_h auth_attempts event_type
| sort -auth_attempts
```

Evidence:

The query filtered failed authentication events where `auth_attempts` was equal to or greater than 5. These events represent brute-force candidates because they show repeated failed SSH authentication attempts.

_time ↕	uid ↕	id.orig_h ↕	id.resp_h ↕	auth_attempts ↕	event_type ↕
2025-04-24 10:20:09.518	SH4068174	10.0.0.19	10.0.1.1	8	Multiple Failed Authentication Attempts
2025-04-24 10:20:09.518	SH5439493	10.0.0.15	10.0.1.7	8	Multiple Failed Authentication Attempts
2025-04-24 10:20:09.518	SH9535038	10.0.0.24	10.0.1.10	8	Multiple Failed Authentication Attempts
2025-04-24 10:20:09.518	SH7403822	10.0.0.18	10.0.1.3	8	Multiple Failed Authentication Attempts
2025-04-24 10:20:09.518	SH4845525	10.0.0.56	10.0.1.9	8	Multiple Failed Authentication Attempts
2025-04-24 10:20:09.518	SH5536175	10.0.0.13	10.0.1.9	8	Multiple Failed Authentication Attempts
2025-04-24 10:20:09.518	SH5348995	10.0.0.57	10.0.1.6	8	Multiple Failed Authentication Attempts

Query 7: Suspicious Source Timeline

Objective: Reconstruct the chronological activity of a suspicious source IP.

```
index=auth_logs id.orig_h="10.0.0.25"
| sort _time
| table _time uid id.orig_h id.orig_p id.resp_h id.resp_p auth_success auth_attempts event_type
```

Evidence:

The timeline query focused on source IP 10.0.0.25 to reconstruct its SSH activity in chronological order. The search returned 78 SSH-related events from this source. The activity included 18 failed SSH login events, 26 multiple failed authentication attempt events, 18 connections without authentication, and 16 successful SSH login events. The failed authentication activity reached a total of 174 authentication attempts, with the highest single event showing 8 attempts. Based on this pattern, 10.0.0.25 was identified as a suspicious source IP that requires monitoring.

Query 8: Alert Search Logic

Objective: Create a reusable search that can be saved as a Splunk alert.

```
index=auth_logs auth_success=false
| where auth_attempts >= 5
| stats max(auth_attempts) as max_attempts count by id.orig_h id.resp_h
| where max_attempts >= 5
| sort -max_attempts
```

Evidence:

Create a Splunk alert for failed SSH authentication events where auth_attempts is greater than or equal to 5.

5. Findings

Finding	Evidence	Severity
Suspicious SSH activity from source IP 10.0.0.25	The timeline query returned 78 SSH-related events from this source IP.	Medium
Repeated failed authentication activity	The activity included 18 failed SSH login events and 26 multiple failed authentication attempt events.	High
Possible brute-force behavior	Failed authentication activity reached 174 total attempts, with the highest single event showing 8 attempts.	High
Successful logins observed from the same source	The source IP also generated 16 successful SSH login events.	Medium
Connections without completed authentication	The timeline included 18 connection events without authentication.	Low

Key Incident Summary

Suspicious Source IP	10.0.0.25
Target Server	10.0.1.10
Maximum Attempts	8
Event Type	Multiple Failed Authentication Attempts
Authentication Result	false
Time Window	2025-04-24 10:20:09 UTC
Analyst Assessment	Source IP 10.0.0.25 was identified as suspicious due to repeated failed SSH authentication activity. The source

	generated multiple failed authentication attempt events, including events with a maximum auth_attempts value of 8. This pattern may indicate brute-force behavior or repeated unauthorized SSH access attempts and should be monitored through Splunk alerting.
--	--

6. Screenshots

Screenshot 1: Data Loaded in Splunk

Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< Back

Next >

Select Source

Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. [Learn More](#)

Selected File: ssh_logs.json

Select File

Drop your data file here

The maximum file upload size is 500 Mb

✓

File Successfully Uploaded

Input Settings

Optionally set additional input parameters for this data input as follows:

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

Constant value

Regular expression on path

Segment in path

Host field value

splunk

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index

auth_logs

Create a new index

Screenshot 2: Event Type Breakdown

event_type	count
Successful SSH Login	612
Failed SSH Login	610
Multiple Failed Authentication Attempts	606
Connection Without Authentication	572

7. Recommendations

Control	Purpose	Priority
Disable direct root SSH login	Reduces impact of credential guessing against privileged accounts.	High
Enforce strong passwords or SSH keys	Reduces likelihood of successful password attacks.	High
Enable MFA where possible	Adds a second factor beyond password authentication.	High
Use Fail2Ban or lockout rules	Blocks repeated failed authentication attempts.	Medium
Monitor SSH logs in Splunk	Improves detection and response visibility.	High
Create a scheduled Splunk alert	Automates detection when thresholds are exceeded.	High

8. Conclusion

This investigation demonstrated how Splunk can be used to analyze SSH authentication logs, identify repeated failed authentication attempts, isolate suspicious source hosts, and build reusable detection logic.

Final Analyst Statement: The investigation identified source IP 10.0.0.25 as a suspicious SSH source based on repeated failed authentication activity. The analysis showed multiple failed authentication attempt events, successful login activity from the same source, and a maximum auth_attempts value of 8. A reusable Splunk alert search was also created to detect similar repeated failed SSH authentication behavior in the future.

9. References

Splunk Search Reference: <https://docs.splunk.com/Documentation/Splunk/latest/SearchReference>

Splunk - Get Data In: <https://help.splunk.com/en/splunk-enterprise/get-started/get-data-in>

MITRE ATT&CK T1110 - Brute Force: <https://attack.mitre.org/techniques/T1110/>

OWASP - Brute Force Attack: https://owasp.org/www-community/attacks/Brute_force_attack

Dataset Source: <https://github.com/0xrajneesh/30-Days-SOC-Challenge->

[Beginner/blob/main/Challenge%234/Day%2318-Splunk%20Basics%3A%20SSH%20Log%20Analysis.md](https://github.com/0xrajneesh/30-Days-SOC-Challenge-/blob/main/Challenge%234/Day%2318-Splunk%20Basics%3A%20SSH%20Log%20Analysis.md)